Drive By Wire

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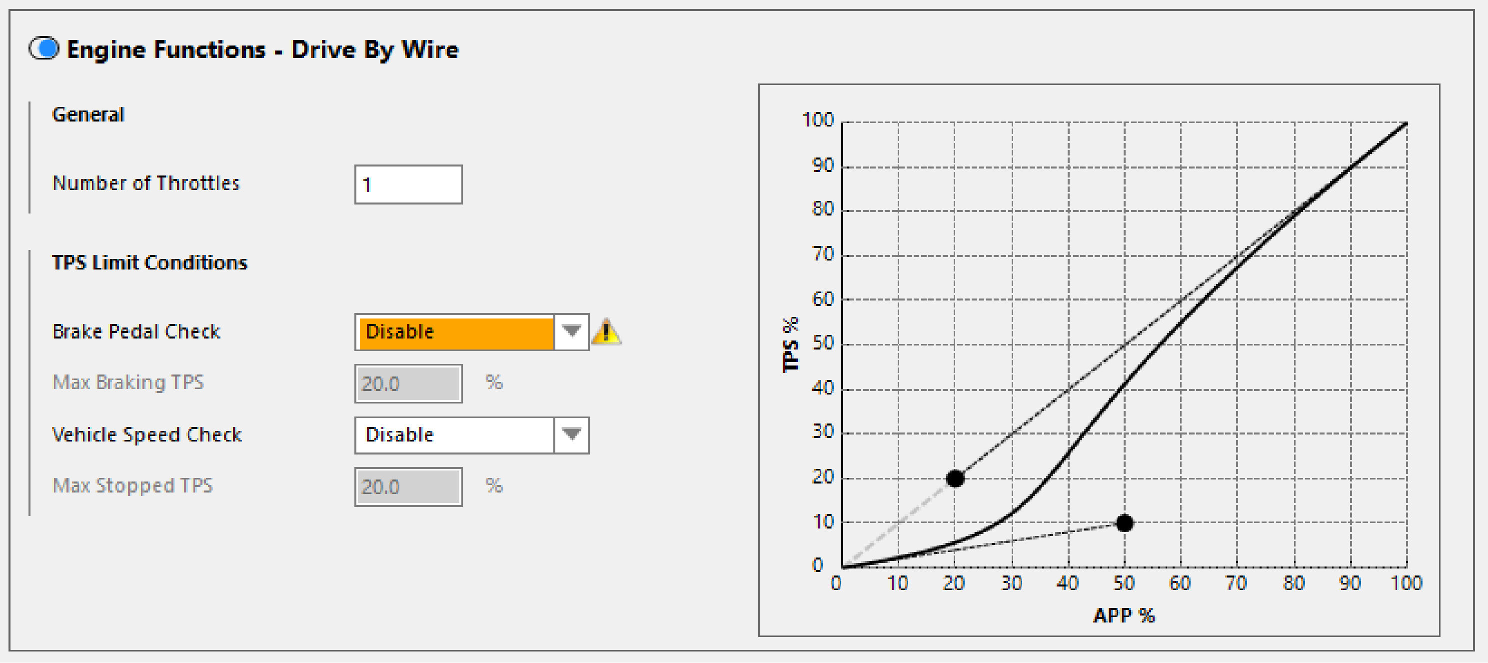
The Drive By Wire (DBW) function allows your ECU to accept an input from an Accelerator Pedal Position (APP) sensor and use this to control the output to a DBW Throttle Body. This type of control allows mapping of the throttle body opening curve, idle control, and lift-off anti-lag control.

A number of safety features are inherent in the design of OEM DBW systems. The APP sensor is required to have two position sensors, and the DBW throttle is also required to have two throttle position sensors (TPS). This requirement is so that the ECU can detect a malfunction in the sensor, and prevent it from commanding the DBW Throttle to a position other than that being sent from the APP. i.e. the throttle cannot open unexpectedly. There are multiple checks being monitored by the ECU to look for fault conditions, and in the event of a failure in the system the output to the DBW motor is cut. This will return the throttle to the rest position which is typically a fixed small amount of throttle opening. The amount of opening at rest is determined by the DBW throttle body design and not the ECU.

Due to the large number of safety systems, there is also a large number of diagnostic channels and information involved with the installation and configuration of a DBW system.



General

Number of Throttles - Select the number of individually controlled DBW Throttles being used (up to 2 on a Nexus R5). If two or more throttles are connected in parallel to one pair of outputs, choose 1.

TPS Limit Conditions

Brake Pedal Check - Enable if a Brake Pedal Switch has been wired and configured. Enabling this function will enable the Max Brake TPS Setting. This setting will override all other commanded throttle values.

Max Braking TPS - The maximum amount of throttle that can be commanded while the brake pedal is depressed. This setting allows for an added level of safety by ensuring the system cannot command full throttle while the brake pedal is depressed. It can also be used as an anti-abuse function by preventing large throttle amounts with the brake is depressed. i.e. burnouts!

Vehicle Speed Check - Enabling this function will enable the Max Stopped TPS setting. This allows a maximum amount of throttle that can be commanded while the vehicle is stationary. A Vehicle Speed Sensor must be configured to use this function.

Max Stopped TPS - The maximum amount of throttle that can be commanded while the vehicle is stationary. Typically this is used to prevent engine abuse.

APP to TPS Relationship Graph

The APP to TPS Relationship Graph allows the user to set the amount of throttle opening that occurs for a given amount of APP opening. The curve is changed by adjusting the position of the two black dots to generate the desired pedal to throttle relationship.